Serendib University Endowment Fund

A conservative USD 10m portfolio and its risk

Team Nexora

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Section 1: Data

Owner: Sameera Ekanayaka **Code:** `src/get_data.py` and `src/data.py`, tested in `tests/test_get_data.py` and `tests/test_data.py` **App page:** Data **Reproduce:** `python -m src.get_data` (download and clean), or `python -m src.get_data --from-raw`

1.1 Sources and period

Data	Source	Series
ETF prices	Yahoo Finance through `yfinance`	Adjusted close for SPY, EFA, EEM, AGG, GLD, XLE, XLK
Risk-free rate	FRED	DTB3, 3-month Treasury bill, annualised percent

We use adjusted close so dividends are in the returns. Without them AGG, which earns most of its return as income, would look far worse than it is.

Prices run from **18 Nov 2004 to 27 Oct 2026** (5,519 trading days). The start is the first day GLD traded, the last of the seven to list, so every day has a price for every ETF. The end date is fixed in `src/config.py` so a rerun gives the same data, and the cleaned files are committed in `data/clean/`. The period covers 2008, March 2020 and 2022, and the script stops if any of them is missing.

1.2 Cleaning

Each step is a small function in `src/get_data.py`. Its row count is saved to `cleaning_report.json`, which the Data page shows.

Step	What it does	Rows removed
Tidy and trim	Sorts dates, drops duplicates, keeps the study period	0
Bad ticks and holidays	Zero or negative prices become missing, days with no trades are dropped	0
Common start	Starts on the first day all seven trade	0
Fill short gaps	Carries the last price over gaps of up to 3 days, stops on longer ones	0 (0 filled)

Yahoo's adjusted data was already clean, so no step changed anything. We kept the steps so a future rerun is safe: a short gap is filled, a long one stops the run instead of being guessed.

FRED does not publish on bond market holidays that are normal stock trading days, so on those **41 days** we carry the last rate forward. The daily rate is annual / 100 / 252.

The script lists any daily move above 15% for a manual check. There are eight, all in October 2008 (EEM, XLE) and March 2020 (XLE). They are real market events, so we keep them. Removing them would hide the tail risk this project is about.

We also flag a possible stale price when an adjusted close is unchanged for three consecutive trading days. There are **no flagged stale-price runs** in the cleaned sample. The check only flags observations for review; it never deletes or changes a price.

We model with daily **log returns** (`load_returns()`), which add up over time. Whenever ETFs are combined into a portfolio or a return becomes USD, we convert to **simple returns**, which add up across assets.

The Augmented Dickey-Fuller test rejects a unit root in every ETF's daily log returns at the 5% level. Lag length is selected by AIC, up to 20 lags. All p-values round below 0.000001; the ADF statistics range from -15.64 to -75.44, well below their 5% critical values of about -2.86. This supports using returns as stationary inputs for the models over the full sample. It does not mean their volatility, correlations or tail behaviour are constant through time.

1.3 What the data looks like

Daily simple returns, Nov 2004 to Oct 2026. Excess kurtosis is 0 for a normal distribution.

ETF	Annual return	Annual volatility	Max drawdown	Worst day	Excess kurtosis
SPY	10.9%	18.8%	-55.2%	-10.9% (16 Mar 2020)	15.5

EFA	6.3%	20.9%	-61.0%	-11.2% (29 Sep 2008)	14.3
EEM	7.1%	27.5%	-66.4%	-16.2% (15 Oct 2008)	17.8
AGG	2.9%	5.2%	-18.4%	-6.8% (10 Oct 2008)	50.5
GLD	10.2%	18.3%	-45.6%	-10.3% (30 Jan 2026)	6.4
XLE	9.3%	29.6%	-71.3%	-20.1% (9 Mar 2020)	11.6
XLK	15.6%	22.6%	-53.0%	-13.8% (16 Mar 2020)	9.8

Three things here shape the rest of the report:

1. **Every ETF has fat tails** (excess kurtosis 6 to 50), so a normal model understates the worst days. That is why Section 3 compares four VaR methods and Section 4 fits EVT to the tail.
2. **AGG is calm but not safe.** It has by far the lowest volatility but the highest kurtosis: long quiet spells broken by sharp falls, as in October 2008 and 2022.
3. **Diversification comes from asset classes, not regions.** The five equity ETFs, XLE and XLK included, move closely together (correlations 0.54 to 0.91). AGG has a correlation of about 0 with every equity ETF, and GLD 0.06 to 0.20.

Section 2: Mandate and portfolio choice

Owner: Imasha Karunathilaka **Code:** `src/portfolio.py` and `src/kpis.py`, tested in `tests/test_mandate.py`, `tests/test_candidates.py`, `tests/test_kpis.py`, `tests/test_tradeoffs.py`, `tests/test_min_es.py`, `tests/test_final_weights.py` and `tests/test_portfolio_page.py` **App page:** Portfolio **Reproduce:** `streamlit run app.py`, then open the Portfolio page. The final weights are in `final_weights()`.

The committee told us that avoiding large losses matters more than making large gains. So we did not look for the best return. We looked for a portfolio whose worst case we can measure and defend. This section sets the rules (the mandate), explains how we built five candidates, and shows why we chose one. Numbers use 5,518 trading days from 19 Nov 2004 to 27 Oct 2026. The average risk free rate over this time is 1.79% a year.

Read this first. The weights were chosen on the same 2004 to 2026 data that we use to measure them. So every return and risk figure in this section is historical and in-sample. We did not test weights chosen on one period against another period.

2.1 The mandate

Decision	Our choice	Reason
Horizon	Long term. Rebalance every quarter in practice.	An endowment lasts for many years.
Return objective	About 7% a year	See below
Style	Long only, no leverage	The fund cannot borrow or short.
One-day 99% VaR limit	1.5% of the fund (USD 150k)	Proposed in Section 8
One-day 99% ES limit	2.0% of the fund (USD 200k)	Proposed in Section 8
Crisis loss limit	15% of the fund (USD 1.5m), measured over the whole crisis window	See below

Return objective. An endowment must pay its spending and keep up with inflation. We set a base nominal return objective of about 7.0% a year using an assumed 4.5% annual spending rate and 2.5% general inflation. We also test two stricter hurdles as policy sensitivities: 7.4% (an assumed 4.9% spending rate plus 2.5% inflation) and 7.8% (an assumed 4.5% spending rate plus 3.3% higher-education cost inflation). These are explicit team assumptions used to assess the return-risk trade-off, not forecasts or model inputs. All model data comes from the approved sources only: prices from yfinance and the risk free rate from FRED. The 7.4% that our chosen portfolio earned in the past is a result, not a target. It clears the 7.0% base objective, matches the 7.4% hurdle, and falls short of the 7.8% hurdle. A conservative mix earns less than endowments that hold private assets, and we show this gap openly in 2.3.

Income. The client also wants income. Our returns use adjusted close prices, so they include the effect of dividends and bond interest, and they show the total return of each ETF. We did not model the cash yield or the yearly spending payments, so we do not state an income figure. The portfolio is 58.85% bonds, which is where most of its income comes from. We treat income as a limitation and not as a result.

Weight limits. These are our own judgement. They keep the fund conservative and stop one ETF from dominating.

ETF	Minimum	Maximum
SPY	0%	40%
EFA	0%	30%
EEM	0%	10%
AGG	30%	60%
GLD	0%	15%
XLE	0%	10%
XLK	0%	15%

Why these limits. The limits are our judgement, not the output of a model. Each one has a reason. Bonds have a 30% floor and a 60% cap. The floor keeps a stable core in a fund that puts losses first. The cap stops the fund from relying on one asset class, which matters because bonds fell together with stocks in 2022. SPY is capped at 40% so that stocks cannot dominate. EFA is capped at 30% to keep some room for other countries. Emerging markets are capped at 10% because they are riskier and we want the fund to stay conservative. Gold is capped at 15% because it is a diversifier and not a source of growth. The two sector funds are capped because they are concentrated bets. Energy gets 10%. Technology gets 15%, a little more, and Section 2.5 shows that lowering this cap barely changes the result. Some limits matter more than others. In Maximum Sharpe, gold and technology sit exactly at their caps and bonds are just below theirs, so the result depends on our choices.

Crisis loss limit. Our first idea was 10% of the fund. We tested it and no candidate meets it in all three historical crises (see 2.3). Even the calmest candidate loses 12.5% in 2022, so a 10% limit cannot be met inside this mandate. We propose 15% (USD 1.5m), the same limit used in Sections 6 and 8. We think this is fair for a conservative investor for four reasons. First, we did not set it to fit our result. The worst historical loss of the final portfolio is 12.6%, which is 2.4 points below the limit. Second, in each crisis the final portfolio lost far less than SPY alone: 5.2% against 36.9% in 2008, 9.9% against 33.4% in March 2020, and 12.6% against 18.2% in 2022. Third, the limit is measured the same way for every portfolio, as the loss over the whole crisis window. Inside 2022 the fall from peak to trough reached 17.2%, which is above 15%, and we say so openly. Fourth, a limit is a trigger for the committee to review the portfolio, not a promise that losses stay below it. The inflation shock (17.2%) is also above the limit and we report it openly. Every candidate except equal weight stays inside 15% in the three historical replays.

2.2 How we built the candidates

The optimized candidates follow the stated weight limits. Equal weight is an unconstrained benchmark and breaches the bond floor and some asset caps. The optimiser stops with an error if it fails, so no wrong result is used quietly.

- **Equal weight.** The closest allowed mix to 1/7 in each ETF. Plain 1/7 breaks the 30% bond floor.
- **Minimum variance.** The calmest allowed mix.
- **Maximum Sharpe.** The best return for each unit of risk, using the 1.79% risk free rate.
- **Risk parity.** The mix where each ETF adds the same share of the total risk. The optimiser reached this exactly: every ETF contributes 14.3%.
- **Minimum ES.** The allowed mix with the lowest historical 95% expected shortfall, solved as a linear programme. It gives the same weights as minimum variance, because the caps on AGG (60%) and GLD (15%) decide the answer.

We also drew the efficient frontier under the same limits. It uses the average yearly return, so it sits a little above the compound returns in the tables. Statistics use log returns. Portfolio returns use simple returns, because log returns do not add across assets.

2.3 The trade-off: return against risk

All numbers are historical and in-sample. Stress losses are measured over the whole crisis window. The inflation shock is our own assumption, not data.

Candidate	Return	Volatility	Max drawdown	1-day ES 99%	2008	Mar 2020	2022	Inflation shock
Equal weight	9.0%	12.8%	-35.4%	3.4%	-19.0%	-21.4%	-7.2%	-18.7%
Minimum variance	6.5%	6.7%	-17.8%	1.7%	-5.6%	-9.6%	-12.5%	-16.0%
Maximum Sharpe	7.4%	7.2%	-18.3%	1.7%	-5.2%	-9.9%	-12.6%	-17.2%
Risk parity	7.0%	8.1%	-23.0%	2.1%	-8.7%	-12.7%	-8.9%	-16.4%
Minimum ES	6.5%	6.7%	-17.8%	1.7%	-5.6%	-9.6%	-12.5%	-16.0%

What the table shows:

- Equal weight earns the most but its worst fall is 35.4%. It also breaks the 15% limit in 2008 and March 2020.
- No candidate keeps every historical crisis under 10%. Minimum variance and Maximum Sharpe both lose about 12.5% in 2022, because stocks and bonds fell together that year. Risk parity loses 12.7% in March 2020.
- The 2022 loss is lower for equal weight and risk parity. This is probably because they hold more energy, which rose that year. It is a one year effect, so we do not treat energy as a hedge.
- Every candidate loses more than 15% in the inflation shock. This is a hypothetical stress and we report it openly.
- Only Maximum Sharpe reaches the 7% target (7.4%). Minimum variance is about half a point short (6.5%).
- Against the two stricter policy sensitivities, 7.4% (higher spending) and 7.8% (higher education inflation), Maximum Sharpe only matches the first and falls short of the second. Only equal weight (9.0%) clears both, and it has the worst losses. This is the price of a conservative mandate.
- The final portfolio's one-day 99% ES is USD 173k (1.73% of the fund), inside the USD 200k limit. Equal weight (3.4%) and risk parity (2.1%) are above the 2.0% ES limit.

2.4 Why we chose Maximum Sharpe

The final weights are SPY 9.6%, AGG 58.9%, GLD 15%, XLK 15% and XLE 1.6%. They are fixed in `final_weights()`, and a test fails if they stop matching the optimiser.

1. **Equal weight is too risky.** The committee puts losses first, and equal weight has twice the drawdown.
2. **Maximum Sharpe and minimum variance lose about the same in the three historical crises.** The gaps are 0.4 points or less (2008: -5.2% against -5.6%, March 2020: -9.9% against -9.6%, 2022: -12.6% against -12.5%).
3. **Maximum Sharpe earns more.** Return is 7.4% against 6.5%. Sharpe is 0.79 against 0.72, and Calmar is 0.41 against 0.37. The Sharpe gap is small, and we do not claim it is a proven difference.
4. **It has a cost.** In the inflation shock Maximum Sharpe loses 17.2%, about USD 120k more than minimum variance.
5. **Its beta to SPY is 0.29.** The information ratio is -0.33. This is expected for a portfolio that is mostly bonds, since it lags SPY in good years. SPY lost 36.9% over the 2008 window, while our portfolio lost 5.2%.

Diversification

The committee asked to see the benefit of diversification. We measure it in two ways, using the full 2004 to 2026 history.

The first way compares risk. If every ETF moved together, the portfolio's volatility would equal the weighted average of the ETF volatilities, which is 11.5%. The portfolio's actual volatility is 7.2%. So mixing the ETFs removes about 37% of the risk, and the diversification ratio is 1.6.

The second way looks at correlations. Bonds (AGG) have a correlation of about zero with the stock ETFs: 0.00 with SPY, 0.00 with XLK, 0.04 with EFA and -0.01 with EEM. Gold is also low (0.07 with SPY and 0.06 with XLK). The stock ETFs, by contrast, move closely together (SPY with XLK 0.91, SPY with EFA 0.88, SPY with XLE 0.70). So the benefit comes from bonds and gold, not from spreading across stock funds. The stock part of the portfolio (SPY, XLK and XLE, about 26% of the money) is not diversified on its own.

Two cautions. These are averages over 22 years. In 2022 stocks and bonds fell together, so the near zero correlation does not hold in every period, and our inflation shock assumes this. Also, a large part of the ratio comes from the 58.85% in bonds, which have low volatility (5.2% a year) to begin with.

Reproduce: the ratio is the weighted average of the asset volatilities from `asset_stats` divided by the portfolio volatility, using `final_weights()`, in `src/portfolio.py`.

Maximum Sharpe against Minimum variance

These two are the closest candidates, so this is the choice a judge will test.

	Minimum variance	Maximum Sharpe

Holdings	SPY 25%, AGG 60%, GLD 15%	SPY 9.6%, AGG 58.9%, GLD 15%, XLK 15%, XLE 1.6%
Annual return	6.5%	7.4%
Annual volatility	6.7%	7.2%
Sharpe	0.72	0.79
Sortino	1.02	1.13
Calmar	0.37	0.41
Max drawdown	-17.8%	-18.3%
1-day ES at 99%	1.7%	1.7%
2008	-5.6%	-5.2%
March 2020	-9.6%	-9.9%
2022	-12.5%	-12.6%
Inflation shock (assumption)	-16.0%	-17.2%

What Maximum Sharpe gives: 0.9 points more return a year, which is what lets it reach the 7% base objective. It also holds four ETFs above 5%, against three.

What it costs: a max drawdown 0.5 points deeper, and an inflation shock loss 1.2 points larger (about USD 120k). In the three historical crises the two are within 0.4 points of each other.

Concentration: Minimum variance gets 51.5% of its variance from SPY alone. Maximum Sharpe gets about 36% of its risk from XLK. Both depend on one ETF, so neither is clearly more diversified.

We do not claim the gap in Sharpe is proven, because both are in-sample. We would switch to Minimum variance if the committee ranks the inflation shock above the 7% objective.

Where the risk comes from. We used Aadhila's risk budget (Section 3), which splits the risk of the final portfolio between the ETFs. Technology is 15% of the money but about 36% of the risk. This is the same under the normal method (36.0%) and historical ES at 99% (36.5%), so it does not depend on the method. The historical VaR attribution is a local approximation around the sample quantile and gives XLK 37.8%; it does not reconcile exactly to portfolio VaR. Bonds are 58.9% of the money but only about 21% of the risk. Gold adds 15.9% of the tail loss, less than its 20.1% share under the normal method, so it helps a little on the worst days. SPY adds 21.7%. The historical numbers use about 55 bad days at 99%, so we read them as rough. **Reproduce:** `risk_budget(returns, final_weights(), alpha=0.99, method='historical')` in `src/risk_budget.py`. **Verdict for the committee:** accept with an adjusted risk limit (Section 8).

2.5 What we tested and what we did not

Lower caps on technology. We reran Maximum Sharpe with a lower cap on XLK.

XLK cap	Return	Sharpe	2022 loss	Inflation shock
15% (chosen)	7.4%	0.786	-12.6%	-17.2%
12%	7.3%	0.775	-12.8%	-17.1%
10%	7.2%	0.767	-12.8%	-17.0%
8%	7.2%	0.758	-12.9%	-16.9%
5%	7.1%	0.745	-13.1%	-16.8%

Reproduce: `xlk_cap_test(returns, rf_annual)` in `src/portfolio.py`

Return and Sharpe fall a little, the 2022 loss gets slightly worse, and the inflation shock improves by only 0.4 points at the lowest cap. The money moves into SPY, which is also stock risk. So we kept 15%.

What we did not do:

- The weights were chosen on the same 2004 to 2026 data used to measure them. The 7.4% return is historical and probably too optimistic. We did not run a test where the weights are chosen on one period and measured on another.
- The headline numbers assume the weights are reset every day, but a real fund rebalances every quarter. We checked this with `rebalancing_comparison`. For the final portfolio the quarterly return is 7.3% against 7.4%, the max drawdown is -18.7% against -18.3%, and the 1-day 99% ES is 1.71% against 1.73%. The difference is small, so we keep the daily figures as the headline. The stress replays in Section 6 already use buy and hold. **Reproduce:** `rebalancing_comparison(returns,`

`final_weights(), rf_annual)` in `src/portfolio.py`.
- We ignore trading costs and fees.
- We measure risk over one day. We did not build a 10 to 20 day view.

2.6 Why a conservative endowment should accept this portfolio

The committee asked what can go wrong, how bad losses could become, and how confident we are. Our answer:

- **What it offers.** Historically 7.4% a year with 7.2% volatility, close to our 7% base objective, with a Sharpe ratio of 0.79.
- **How bad it got.** In the three historical crises it lost 5.2%, 9.9% and 12.6% over the crisis window (at most USD 1.26m on USD 10m), against 36.9%, 33.4% and 18.2% for SPY alone. The one day 99% expected shortfall is 1.73% of the fund (USD 173k), inside the USD 200k limit.
- **What remains.** Stocks and bonds can fall together, as in 2022. Our own inflation shock costs 17.2% (USD 1.72m), above the 15% limit. Technology causes about 36% of the risk with 15% of the money.
- **How confident we are.** Less than the numbers suggest. The weights were chosen on the same 2004 to 2026 data, trading costs are not included, and the quarterly rebalancing check shows a small drop in return.
- **Our request.** Accept the portfolio with a crisis loss limit of 15%, and review it if the inflation shock or the technology risk becomes a concern.

2.7 Other risks we considered

These come from our own tests. They are not forecasts.

- **Interest rates.** Bonds are 58.85% of the fund. A fast rise in yields is the biggest single risk, as in 2022. Section 6 assumes bonds fall 15% in our inflation shock.
- **Inflation.** Our inflation shock costs 17.2% (USD 1.72m), above the 15% limit. Section 8 says we would move money from technology into energy if inflation persists.
- **Stock and bond correlation.** The portfolio relies on bonds and gold to offset stocks. If stocks and bonds move together for a long time, that protection is weaker. This is why 2022 is our worst historical replay.
- **Concentration.** Technology is 15% of the money and about 36% of the portfolio's risk, so a fall in technology shares matters more than its weight suggests.
- **Model risk.** Our risk estimates come from history. Section 5 shows that breaches cluster in crises for every model, so the limits are monitoring triggers and not guarantees.

Section 3: VaR, Expected Shortfall and risk budget

Owner: Aadhila Anees **Code:** `src/var_es.py` and `src/risk_budget.py`, tested in `tests/test_var_es.py` and `tests/test_risk_budget.py` **App page:** Risk dashboard

All numbers are for the final portfolio (`final_weights()`), one day ahead, on the USD 10m fund, using daily returns from Nov 2004 to Oct 2026.

3.1 Four ways to measure a bad day

VaR at 99% is the loss we expect to beat on only 1 day in 100. ES is the average loss on those worst days, so it says how deep the tail goes once VaR is breached. We measure both four ways:

- **Historical simulation** reads the quantile straight from the 5,518 real daily returns. It assumes this sample represents future risk and cannot see a loss worse than history, but it imposes no distributional form.
- **Parametric normal** fits a mean and a standard deviation.
- **Parametric Student t** fits the tail thickness as well. On our portfolio the fitted degrees of freedom is about 3.3, which is a very fat tail.
- **Monte Carlo** draws 100,000 returns from the fitted Student t with a fixed seed (42), so every run gives the same number.

The normal and Student t estimates include the sample mean as one-day drift, which is the implementation default. We also reran them with zero drift, a more conservative short-horizon assumption. At 99%, normal VaR/ES move from 1.02%/1.17% to 1.05%/1.20%, and Student t VaR/ES move from 1.16%/1.73% to 1.20%/1.77%. The largest change is 0.04 percentage points, so the limit verdict does not depend on this

choice.

3.2 Results

Level	Method	VaR	VaR (USD)	ES	ES (USD)
95%	Historical	0.66%	66,200	1.02%	102,300
95%	Normal	0.71%	71,300	0.90%	90,100
95%	Student t	0.61%	60,600	0.98%	98,100
95%	Monte Carlo	0.61%	60,600	0.97%	96,900
99%	Historical	1.16%	116,300	1.73%	173,400
99%	Normal	1.02%	102,000	1.17%	117,300
99%	Student t	1.16%	115,700	1.73%	172,600
99%	Monte Carlo	1.15%	115,200	1.67%	167,400

3.3 Why the models disagree

At 95% the normal model gives the highest VaR, because real returns are more peaked in the middle than a bell curve. At 99% it flips: the normal model gives the lowest VaR and by far the lowest ES. Its 99% ES of USD 117k is **USD 56k below** what history shows (USD 173k). The returns have fat tails (Section 1), and a normal curve gives almost no chance to the 4 and 5 standard deviation days that happened in 2008 and 2020.

The Student t and Monte Carlo numbers land within 4% of history at 99%, because they fit the tail. This is why the memo (Section 8) monitors the daily limits with Student t, and why Section 5 finds the normal model breaches twice as often as it should.

Why ES as well as VaR. VaR only says where the tail starts. ES says what an average bad day in the tail costs, here USD 173k against a VaR of USD 116k. ES is also coherent: the ES of a combined portfolio is never more than the sum of its parts, so it never punishes diversification. VaR does not have this property (Artzner et al., 1999).

3.4 Risk budget: where the tail risk comes from

Because ES scales with the weights, each asset's share can be split exactly by Euler allocation: weight times marginal ES, summed over assets, gives the portfolio ES. The table uses historical ES at 99%. Historical VaR is a sample quantile and is not smooth, so its component allocation is only a local estimate from returns around the quantile; unlike ES, it does not reconcile exactly to portfolio VaR.

ETF	Weight	Marginal ES	Contribution (USD)	Share of ES
SPY	9.55%	3.94%	37,585	21.7%
AGG	58.85%	0.62%	36,497	21.0%
GLD	15.00%	1.84%	27,658	15.9%
XLE	1.60%	5.20%	8,324	4.8%
XLK	15.00%	4.22%	63,366	36.5%
Total	**100%**		**173,429**	**100%**

EFA and EEM have zero weight, so they add nothing. The parametric (normal) split puts XLK at 36% too, so the main finding does not depend on the method.

What this tells the committee:

1. **Capital is not risk.** AGG is 59% of the money but only 21% of the tail risk. XLK is 15% of the money but **37% of the tail risk**, the largest single source.
2. **XLK is the clearest position to monitor.** Its marginal ES is almost seven times AGG's, so a direct cut lowers technology exposure. It is not a complete fix by itself: the lower-cap test in Section 2 shifts most of the money into SPY and improves the inflation shock by only 0.4 percentage points when the cap falls from 15% to 5%.
3. **GLD earns its place.** It is as large as XLK but adds well under half as much tail risk, because it moves little with stocks.

Section 4: Extreme Value Theory (EVT) and Tail Risk

Owner: Praveen Madawalage
Mandate: USD 10m Serendib University Endowment Fund | **Sample:** 19 Nov 2004 to 27 Oct 2026 (5,518 days)
Code: `src/evt.py`, tested in `tests/test_evt.py` | **App page:** Extreme Value Theory

For an endowment, the core fiduciary objective is avoiding catastrophic losses. Gaussian models assume extreme crashes are virtually impossible, while Historical Simulation cannot extrapolate beyond past data. Peaks-Over-Threshold (POT) Extreme Value Theory overcomes both by modeling tail exceedances directly.

4.1 Theoretical Foundation: Peaks-Over-Threshold (POT)

Under the Pickands-Balkema-de Haan Theorem, exceedances $Y = X - u$ above threshold u converge to the **Generalized Pareto Distribution (GPD)**:

$G_{\xi,\sigma}(y) = 1 - (1 + \xi y/\sigma)^{-1/\xi}$, for $\xi \neq 0$ and $y \geq 0$ with $1 + \xi y/\sigma > 0$.

where ξ is the shape parameter and $\sigma > 0$ is the scale parameter:

- **$\xi > 0$ (Fréchet / Heavy-tailed):** Tails decay as a power law $P(X > x) \sim x^{-1/\xi}$. Most financial return series exhibit fat tails and crash risk.
- **$\xi = 0$ (Gumbel):** Unbounded exponential tail.
- **$\xi < 0$ (Weibull):** Bounded tail with a finite endpoint.

4.2 Threshold Selection and Diagnostic Evidence

Selecting threshold u balances an empirical bias-variance trade-off: setting u too low violates GPD convergence, while setting u too high inflates MLE variance.

We selected the **90th percentile of losses** as the threshold ($u = 0.45\%$, giving $N_u = 552$ tail observations across 5,518 trading days) after reviewing three diagnostic plots:

1. **Mean Excess Plot:** Empirical mean excess $e(u) = E[X - u \mid X > u]$ slopes upward approximately linearly above $u \approx 0.45\%$, which is consistent with the GPD assumption.
2. **Parameter Stability Plot:** Across thresholds from 75% to 98%, fitted shape ξ stays within about 2 standard errors (± 2 SE $\approx \pm 0.10$, shape_se ≈ 0.050), and modified scale $\sigma^*(u) = \sigma(u) - \xi u$ is broadly stable above 0.45%.
3. **Q-Q Plot:** Theoretical GPD quantiles track empirical exceedances reasonably along the 45-degree line.

The diagnostics are broadly consistent with the selected threshold, although threshold uncertainty and finite-sample error remain. The threshold is not automatically optimized; it is a percentile-based choice whose adequacy is assessed visually.

Metric	Equal Weight	Minimum Variance	Maximum Sharpe (Final)	Risk Parity
90th Percentile Rule	0.78%	0.41%	**0.45%**	0.49%
95th Percentile Rule	1.14%	0.60%	**0.66%**	0.72%
Selected Threshold (u)	0.78%	0.41%	**0.45%**	0.49%
Exceedance Count (N_u)	552 days	552 days	**552 days**	552 days
Fitted Shape (ξ)	0.2352	0.1921	**0.1843**	0.2053

On the final Maximum Sharpe portfolio, MLE estimation yields $\hat{\xi} = 0.1843$ and $\hat{\sigma} = 0.002696$, confirming heavy-tailed Fréchet behavior.

4.3 EVT VaR and Expected Shortfall

With exceedance fraction $\hat{p}_u = N_u/N = 10.0\%$, closed-form EVT Value at Risk and Expected Shortfall are:

$VaR_\alpha = u + (\hat{\sigma}/\hat{\xi})[(\hat{p}_u/(1 - \alpha))^{\hat{\xi}} - 1]$; $ES_\alpha = (VaR_\alpha + \hat{\sigma} - \hat{\xi}u)/(1 - \hat{\xi})$.

The ES expression requires $\hat{\xi} < 1$. At $\hat{\xi} = 0$, the continuous limit is $VaR_\alpha = u + \hat{\sigma} \log(\hat{p}_u/(1 - \alpha))$ and $ES_\alpha = VaR_\alpha + \hat{\sigma}$. For $\hat{\xi} \geq 1$, the fitted GPD has infinite theoretical ES.

The table compares 1-day tail risk metrics on the **USD 10m Maximum Sharpe Portfolio**:

Level	Methodology	1-Day VaR (%)	1-Day VaR ($)	1-Day ES (%)	1-Day ES ($)	Tail Gap (ES - VaR)
95%	Normal (Gaussian)	0.71%	$71,300	0.90%	$90,100	$18,800
	Historical Simulation	0.66%	$66,200	1.02%	$102,300	$36,100
	EVT (POT / GPD)	**0.65%**	**$64,637**	**1.02%**	**$102,200**	**$37,563**
99%	Normal (Gaussian)	1.02%	$102,000	1.17%	$117,300	$15,300
	Historical Simulation	1.16%	$116,300	1.73%	$173,400	$57,100
	Student-t	1.16%	$115,700	1.73%	$172,600	$56,900
	EVT (POT / GPD)	**1.22%**	**$122,044**	**1.73%**	**$172,578**	**$50,534**

4.4 Practical Committee Takeaways

1. **Gaussian Blindspot:** At 99% confidence, Normal VaR predicts a daily loss of $102k, while EVT projects $122k. More critically, Normal ES ($117k) understates tail loss by **$55,278 (-32.0%)**. Sizing capital on Gaussian models leaves the fund dangerously under-buffered.
2. **Body vs. Tail Alignment:** At 95% confidence, EVT ES ($102,200) matches Historical Simulation ($102,300) almost to the dollar. This agreement is a consistency check, not evidence of improved precision; tail estimates remain sensitive to the threshold and sample.
3. **Tail Gap & Capital Policy:** At 99%, the EVT Tail Gap is **$50,534**: when breaches occur, losses extend 41.4% deeper than VaR. This supports the Section 8 policy: use **Expected Shortfall ($173k)** for capital sizing and adopt the proposed **USD 200,000 daily ES limit**.

Section 5: Backtesting

Owner: Tharindu Dhanushka **Code:** `src/backtest.py`, tested in `tests/test_backtest.py` **App page:** Backtesting **Reproduce:** `python -m src.backtest`

A 99% VaR model promises a loss bigger than the forecast on about 1 day in 100. We check that promise on days the model has not seen, for the final portfolio (`final_weights()`, Section 8), on data to 27 Oct 2026. We ask two questions: are there the right number of breaches (coverage), and do they arrive one at a time rather than in clusters (independence)?

5.1 Set-up

- **Out of sample, conditional on fixed weights.** Day t uses only the prior 500 returns. The weights were selected on the full 2004 to 2026 sample, so this tests the risk forecasts, not the portfolio-selection rule.
- **500-day window (about two years).** It is above the Basel one-year minimum, leaves five observations in the 1% tail, and puts the first forecast on 14 Nov 2006, so all of 2008 is tested out of sample. The app also offers 250 and 1000 days as a robustness check.
- **Refitting.** Historical and normal VaR are refit daily. Student t, Monte Carlo (20,000 draws, fixed seed) and EVT are refit every 21 trading days and held in between, since each fit takes 0.1 to 0.3 seconds.
- **Sample.** 5,018 forecast days, 14 Nov 2006 to 27 Oct 2026.

5.2 Tests

- **Kupiec** compares x breaches in n days with the target rate p = 1 - alpha: LR_pof = -2 ln[(1-p)^(n-x) p^x / ((1-x/n)^(n-x) (x/n)^x)], chi-squared(1).
- **Christoffersen independence** compares the chance of a breach after a breach (pi_11) with the chance after a quiet day (pi_01), chi-squared(1). **Conditional coverage** adds the two statistics, chi-squared(2).
- **Basel traffic light**, defined for 99% VaR over 250 days: 0 to 4 breaches green, 5 to 9 yellow, 10 or more red. We apply it to the last 250 days. For the whole range we use the binomial rule the zones come from (green below 95% cumulative probability, red from 99.99%); a test checks it reproduces the official table. At 90% and 95% the light does not apply and the app says so.
- **ES check (extra).** The Acerbi and Szekely Z2 statistic is near 0 when breach losses match the forecast ES and positive when ES is too low. Its cut-offs (0.7 yellow, 1.8 red) are for 97.5%, so we treat them as indicative.

5.3 Results

At 99% (50.2 breaches expected)

Model	Breaches	Kupiec p	Indep. p	Whole range	Last 250 days	ES Z2
Historical simulation	67	0.023	<0.001	Yellow	Green (4)	+0.51
Parametric (normal)	105	<0.001	<0.001	Red	Yellow (6)	+1.75
Parametric (Student t)	41	0.179	<0.001	Green	Green (4)	-0.08
Monte Carlo (Student t)	41	0.179	<0.001	Green	Green (4)	-0.04
EVT (GPD)	56	0.418	<0.001	Green	Green (4)	+0.60

Conditional coverage is rejected for every model ($p < 0.001$).

At 95% (250.9 expected), breaches and Kupiec p are: historical 277 (0.096), normal 270 (0.221), Student t 257 (0.694), Monte Carlo 261 (0.516), EVT 284 (0.036). Independence is rejected for every model ($p < 0.001$).

Breaches at 99% by period

Period	Days	Expected	Historical	Normal	Student t	EVT
2008 financial crisis	146	1.5	12	16	15	14
March 2020 Covid crash	24	0.2	7	8	5	7
2022 rate hikes	251	2.5	5	12	3	4
Calm days	4,597	46.0	43	69	18	31

5.4 What this means

1. **The normal model fails at 99%:** 105 breaches against 50, red over the whole range, and breaches deeper than it expects (Z2 = +1.75). This is the fat-tail problem from Section 3, confirmed in rolling forecasts conditional on the fixed portfolio weights. Its Kupiec test passes at 95%, so a 95% coverage test alone would hide it; independence is still rejected.
2. **Student t, Monte Carlo and EVT pass Kupiec at 99%.** EVT is closest (56 against 50, $p = 0.42$). Student t is right only on average: too cautious in calm years (18 against 46) and too slow in 2008 (15 in 146 days). Monte Carlo draws from the same fitted Student t, so it gives the same count.
3. **Historical VaR is breached a little too often** (67 against 50, $p = 0.02$), which is why Section 8 monitors the daily limit with Student t or EVT instead.
4. **Every model fails independence.** Breaches bunch in 2008 and March 2020 because a fixed 500-day window reacts slowly when volatility jumps. This is the main finding: it points to a conditional model (GARCH-filtered HS or GARCH-EVT) as the next step, and it is why the stress tests in Section 6 carry as much weight as VaR.
5. **No red zone.** Over the last 250 days every model except the normal one is green, so the memo's trigger for changing our mind has not been hit.

5.5 Limitations

- **Low power.** At 99% over 250 days only 2.5 breaches are expected, so one year can rarely reject a bad model. We test over 20 years and by period; the short stress periods give only rough p-values.
- **Held forecasts.** Monthly refits keep a forecast fixed for up to 20 days.
- **Fixed weights.** The portfolio selection is in sample. The risk forecasts use only prior returns. This section rebalances daily; Section 6 uses buy and hold.

Section 6: Stress tests

Owner: Tharindu Dhanushka **Code:** `src/stress.py`, tested in `tests/test_stress.py` **App page:** Stress tests **Reproduce:** `python -m src.stress`

VaR describes a normal bad day, and Section 5 shows every model lets breaches pile up when a crisis starts. Stress tests answer the committee directly: if 2008 happened again, how much of the USD 10m would we lose? Numbers are for the final portfolio (`final_weights()`, Section 8) on data to 27 Oct 2026.

6.1 Method

- **Historical replays** of three periods from `src/config.py`: 2008 (1 Sep 2008 to 31 Mar 2009), the Covid crash (19 Feb to 23 Mar 2020) and the 2022 rate hikes (calendar year). We buy at the close before the first day and hold, so weights drift and nothing is rebalanced mid-crisis, as for a fund that

rebalances quarterly. ETF contributions add up exactly to the total.
- **Our own scenario (an assumption, not data):** an inflation surprise lifts yields about 250bp, so stocks and bonds fall together. This is the stock-bond hedge failure we found in round 1. The shocks can be edited in the app.

ETF	Shock	Reason
SPY, EFA	-25%	Higher real rates cut valuations, about 1.4 times 2022
EEM	-30%	Strong dollar and capital outflows
AGG	-15%	250bp rise in yields times a duration of about 6
GLD	-5%	Rising real yields, so no hedge
XLE	0%	Rose 64% in 2022; we assume no help
XLK	-35%	Long-duration tech falls most when rates rise

- **Limit.** The crisis loss limit is 15% of the fund (USD 1.5m), measured over the whole crisis window (Section 8). The reverse stress test asks how many times as severe our scenario must be to reach it.

6.2 Results

Final portfolio

Scenario	Return	P&L (USD)	Max drawdown	Worst day	SPY
2008 financial crisis	-5.2%	-517,024	-14.6%	-5.86% (10 Oct 2008)	-36.9%
March 2020 Covid crash	-9.9%	-989,191	-12.8%	-5.28% (12 Mar 2020)	-33.4%
2022 rate hikes	-12.6%	-1,264,453	-17.2%	-2.45% (13 Jun 2022)	-18.2%
Inflation shock (assumption)	-17.2%	-1,721,500			-25.0%

Contribution by ETF, in points of return (EFA and EEM have zero weight)

Scenario	SPY	AGG	GLD	XLE	XLK
2008 financial crisis	-3.5	+2.0	+1.6	-0.7	-4.6
March 2020 Covid crash	-3.2	-0.8	-0.5	-0.9	-4.6
2022 rate hikes	-1.7	-7.7	-0.1	+1.0	-4.2
Inflation shock	-2.4	-8.8	-0.8	0.0	-5.3

Every candidate

Scenario	Equal weight	Min. variance	Max. Sharpe (final)	Risk parity
2008 financial crisis	-19.0%	-5.6%	-5.2%	-8.7%
March 2020 Covid crash	-21.4%	-9.6%	-9.9%	-12.7%
2022 rate hikes	-7.2%	-12.5%	-12.6%	-8.9%
Inflation shock	-18.8%	-16.0%	-17.2%	-16.4%
Reverse stress multiple at 15%	0.80	0.94	0.87	0.91

6.3 What this means

1. **Diversification works in equity crashes.** In 2008 the portfolio lost 5.2% while SPY lost 36.9%, as AGG (+2.0 points) and GLD (+1.6) offset most of the equity losses.
2. **It fails when rates rise.** In 2022 AGG, at 59% of the fund, cost 7.7 of the 12.6 points. So the bond-heavy candidates (Maximum Sharpe, Minimum variance) lose least in 2008 and 2020 but most in 2022. XLE, up 64% that year, contributed positively in that episode; we hold 1.6% of it and do not assume this repeats.
3. **Every replay ends inside 15%, but 2022 does not stay inside it.** The deepest fall within the year was 17.2% (to 20 Oct 2022). That is why the limit is measured over the whole crisis window, not peak to trough.
4. **Our inflation scenario breaks 15% for every candidate.** The portfolio loses 17.2% (USD 1.72m), of which AGG is USD 883k and XLK USD 525k. A scenario 0.87 times as severe already reaches the limit.
5. **Short crises are not small.** March 2020 lasted 24 trading days but cost almost twice our 2008 window (-9.9% against -5.2%), because AGG and GLD fell too in the dash for cash.

The largest single risk is rates through AGG, then XLK. Section 8 sets out what we would change if inflation picks up.

6.4 Limitations

- **The 2008 window starts on 1 Sep 2008**, so it misses the fall from the October 2007 peak and understates the full bear market.
- **History will not repeat exactly**, which is why we add our own scenario. Its shocks are judgement, applied instantly (no path or drawdown), and the reverse stress test assumes they scale in proportion.
- **The AGG shock is approximate.** The calculation $-D \times \Delta y$ assumes a parallel 250bp yield shift and duration 6. It ignores convexity, spread and duration changes, and the rate path; actual losses can be higher or lower.
- **No transaction costs or liquidity effects.** ETF spreads widened sharply in March 2020, so selling in a crisis would cost more than shown.

Section 7: Assumptions and Limitations

Owner: Praveen Madawalage
Mandate: USD 10m Serendib University Endowment Fund | **Status:** Model Risk Governance

Quantitative risk models are simplifications of non-stationary financial markets. The Serendib Endowment mandate instructs that avoiding large losses takes priority over maximizing returns. Fulfilling this duty requires transparency regarding model limits and how we govern them.

7.1 Layer-by-Layer Assumptions and Mitigations

1. Portfolio Construction (Layers 0 & 1)

- **Costless Daily Rebalancing:** Models assume continuous rebalancing without transaction friction. *Mitigation:* The endowment implements calendar-quarter rebalancing to target weights.
- **Stationarity of Return Moments:** Covariances over 2004-2026 are assumed representative. Macro regime shifts alter distributions.
  Mitigation: We enforce strict structural mandate bounds: a 30% floor on US Aggregate Bonds (AGG) and single-asset caps (GLD 15%, XLK 15%, XLE 10%).

2. Data Pipeline & Cleaning (Layer 1)

- **Sample Horizon (Nov 2004 to Oct 2026):** Starting on 19 Nov 2004 (GLD inception) omits the 2000-2002 dot-com crash.
  Mitigation: The 22-year common sample (5,518 days) spans three major crises: 2008 GFC, March 2020 COVID, and 2022 rate hikes.
- **Calendar Alignment:** Forward-filling missing prices up to 3 days could dampen volatility.
  Mitigation: The audit (`data/clean/cleaning_report.json`) confirms 0 days required fill across our liquid ETF universe.

3. Tail Risk Modeling (Layers 2, 3 & 4)

- **Distributional Assumptions:** Normal models assume thin tails, severely understating tail losses (Normal 99% ES is USD 117k vs. USD 173k Historical/EVT, a USD 56k blindspot). Student-t assumes tail symmetry.
  Mitigation: We mandate Expected Shortfall (coherent, subadditive) as the binding constraint.
- **Square-Root-of-Time Scaling ($\sqrt{T}$):** Extrapolating 1-day risk to multi-day horizons assumes i.i.d. returns, ignoring volatility clustering. *Mitigation:* Multi-week reserves are calibrated directly against empirical crisis drawdowns.
- **EVT Asymptotic Validity:** GPD is an asymptotic limit theorem ($n \to \infty$, $u \to \infty$). *Mitigation:* Threshold $u = 0.45\%$ (90th percentile) provides 552 observations, supported by Mean Excess linearity, parameter stability within ±2 SE, and Q-Q diagnostics.

4. Backtesting and Validation (Layer 5)

- **Binomial Test Power in Finite Samples:** At 99% confidence over 250 days, expected breaches are only 2.5 days, giving Kupiec tests low power to detect underestimation (Type II error).

Mitigation: We backtest over the full 22-year history (5,518 days; with the 500-day estimation window, the out-of-sample forecast count is approximately 5,018 days), combine coverage with independence tests, and monitor Basel traffic lights.

5. Stress Testing and Macro Scenarios (Decision Layer)

- **Crisis Repeatability & Hedge Breakdown:** Past drawdowns may not repeat. In 2022, stocks and bonds fell concurrently.
  Mitigation: We test a synthetic Stagflation Shock. XLE performed well during the 2022 inflation episode (+64%), but we do not assume that result will repeat (Section 2 treats one-year energy performance as unreliable). The inflation scenario assigns XLE a 0% shock, reflecting our conservative stance. Gold (GLD, -0.8% in 2022) is resilient in equity crashes but constrained by real rates. Protocol: rotate XLK into XLE if inflation accelerates.

7.2 Governance Checklist for Investment Committee

Risk Dimension	Model Vulnerability	Institutional Defense / Guardrail
Execution Drag	Transaction friction from turnover	Calendar-quarter rebalancing to target weights
Model Misspecification	Gaussian tail undercounting	Mandatory use of EVT GPD and Expected Shortfall for sizing
Hedge Breakdown	Stock-bond correlation flips positive	Dedicated inflation stress test; protocol to rotate XLK into XLE
Horizon Extrapolation	√T scaling underestimates clustering	Sizing multi-week liquidity buffers on empirical stress drawdowns

Section 8: Committee memo

Decision requested: Conditional acceptance of Maximum Sharpe for the USD 10m endowment, with a 15% crisis-window monitoring limit. Committee approval of the disclosed inflation-stress exception is required; this portfolio does not satisfy a hard 15% ceiling across all scenarios.

SPY	EFA	EEM	AGG	GLD	XLE	XLK
9.55%	0%	0%	58.85%	15%	1.60%	15%

Rationale and trade-off

The portfolio historically returned 7.4% annually with 7.2% volatility, Sharpe 0.79 and maximum drawdown -18.3%. Minimum variance returned 6.5% with -17.8% maximum drawdown. The extra return supports our assumed 7.0% objective (4.5% spending plus 2.5% inflation). It approximately meets the 7.4% spending sensitivity and misses the 7.8% inflation sensitivity. These are team assumptions, not forecasts. Income payments and trading costs are not separately modeled.

Measured downside and proposed limits

Measure	Portfolio	Proposed limit
One-day 99% Student t VaR	1.16%, USD 116k	1.5%, USD 150k
One-day 99% Student t ES	1.73%, USD 173k	2.0%, USD 200k
Worst historical crisis-window loss	12.6%, 2022	15%, USD 1.5m
Hypothetical inflation loss	17.2%, USD 1.72m	Exceeds 15%

Historical crisis losses were 5.2% in 2008 and 9.9% in March 2020. The 2022 peak-to-trough loss was 17.2%; the 15% limit measures the full crisis window, not drawdown. Almost 59% in bonds creates rate exposure, while technology contributes about 37% of historical ES despite a 15% capital weight.

We prefer Maximum Sharpe for its return advantage with similar historical crisis losses to Minimum variance. Neither meets the hypothetical 15% limit. If that ceiling is mandatory, do not approve these weights: require a revised allocation and rerun every risk test. Disclosure alone does not cure a breach.

Monitoring and conditions

Monitor Student t and EVT daily risk, stock-bond correlation, technology concentration and inflation stress. A shift of about 6.5 percentage points from XLK to XLE reduces the specified inflation shock below 15%, but depends on the assumed XLE shock and requires fresh return/risk validation before adoption. Use Minimum variance if lower historical drawdown takes priority over the return objective.

Weights were selected in sample. Risk forecasts use prior observations conditional on those fixed weights. Student t and EVT pass 99% coverage, but all models fail breach independence: losses cluster. Daily rebalancing underlies headline numbers; quarterly rebalancing gives 7.3% return and -18.7% drawdown. Limits are review triggers, not guarantees. Review the allocation on a limit breach, persistent positive stock-bond correlation or a 99%/250-day Basel red signal.

AI disclosure

Each person adds a row for any AI tool they used on their part. The same table goes into the report.

Member	Tool	Purpose	How we validated the output	Parts that were AI-assisted
Praveen Madawalage	Claude / Gemini Coding Assistant	Mathematical implementation of GPD POT formulas, Plotly visualizations, Streamlit test scaffolding	Unit testing against known synthetic GPD distributions (test_evt.py), manual formula verification against Pickands-Balkema-de Haan theorem, verification of monotonicity and subadditivity	Mathematical draft of closed-form POT VaR/ES in `src/evt.py`, interactive Plotly chart scaffolding in `app/pages/4_evt.py`
Imasha Karunathilaka	Claude (Anthropic)	Step by step coding help and drafting of report section 2	Ran every unit test, ruff and black locally and in the CI. Checked the return-hurdle arithmetic against the stated committee assumptions and compared the report numbers with the app output.	Draft code for the optimisers, KPI functions, minimum ES solver, XLK cap test and the Portfolio page (`src/portfolio.py`, `src/kpis.py`, `app/pages/2_portfolio.py`) and their tests. First draft of `docs/report/02_mandate_portfolio.md`
Sameera Ekanayaka	OpenAI Codex; Claude (Anthropic)	Repository audit, coding assistance, research support, data-compliance implementation, final integration, and drafting PR summaries and commit messages	Reviewed changes against the competition brief and approved-source rule; ran the full test, lint and format checks; matched generated ADF, portfolio, risk and stress results to the cleaned data and application; manually reviewed PR and commit text	Draft implementation and tests for ADF and stale-price checks; Data page and Section 1 updates; consistency edits to the committee memo and tear sheet; research notes; PR summaries and commit-message wording
Aadhila Anees	Google Antigravity; ChatGPT	Coding assistance and maintaining context across the VaR, ES and risk-budget work	Recomputed VaR/ES with and without drift; reconciled component results against portfolio totals; ran the relevant tests and compared the report figures with the application	Draft code and explanations for VaR, ES, risk attribution and Section 3 wording
Tharindu Dhanushka	Claude; Claude Code; OpenAI Codex	Understanding the competition context, planning, and coding assistance	Ran the backtesting and stress-test suites; checked breach counts, test statistics and scenario losses against the application and report	Planning notes and draft code for VaR backtesting, validation displays, historical crisis replays and hypothetical stress tests